

Problem: Tangent Parallel Perpendicular Diameter

Let ABC be a triangle with circumcircle ω . Point D lies on side BC .

Draw tangents to ω at B and C , intersecting at M .

Through D , draw lines parallel to MC and MB , meeting CA and AB at E and F , respectively.

Prove that the perpendicular to BC at D is a diameter of the circumcircle of $\triangle AFE$.